HR Analytics Dashboard – Business Requirements Document (BRD)

Project: HR Analytics Final Project (Power BI) **Date:** 30 October 2026 **Prepared by:** Abdullah Ahmed

1. EXECUTIVE SUMMARY

The organization lacks a single, trusted view of its workforce. This project delivers a three-page Power BI report (Workforce Overview, Demographics, Attrition) on a star-schema model so HR and executives can monitor workforce size and cost, understand its composition, and see where and why people leave.

2. BUSINESS CONTEXT AND PROBLEM STATEMENT

Background: HR data is held in a central HR fact table (one record per employee) linked to dimensions for department, job role, location, education, marital status, employment status, employee demographics and date. All monetary values are in Egyptian Pounds (EGP).

Problems to solve:

- No consolidated view of active headcount, payroll and workload.
- Attrition is not measured consistently (voluntary vs. involuntary, by department or role).
- Gender pay gap, age structure and education profile are not visible to leadership.
- Manual reporting is slow and error-prone, and definitions differ between teams.

3. BUSINESS OBJECTIVES AND SUCCESS CRITERIA

ID	Objective	Success measure
BO-1	Provide an accurate view of workforce size, cost and productivity	Headcount and payroll reconcile 100% to source data
BO-2	Expose workforce demographics and pay equity	Gender, age, education, marital and location splits available with one click
BO-3	Quantify and explain attrition	Attrition rate, voluntary/involuntary split and drivers shown by any dimension
BO-4	Standardise HR KPI definitions	One documented definition per KPI, implemented once as a DAX measure
BO-5	Enable self-service analysis	Users filter by department, role, location, tenure, age and salary band without IT help

4. SCOPE

4.1 In scope

- Three report pages: Workforce Overview, Demographics, Attrition.
- Star-schema semantic model: Fact_HR plus dimension tables.
- Calculated columns and DAX measures organized in display folders.
- Dynamic page titles, benchmark comparisons and conditional-formatting measures.

4.2 Out of scope

- Real-time or streaming data; payroll processing; recruitment pipeline (applicants, offers).
- Predictive attrition modelling (listed as a future enhancement).
- Row-level security configuration beyond standard department access.

5. STAKEHOLDERS

Stakeholder	Role	Business Interest
Head of HR	Business Owner / Approver	Strategic workforce insight, retention tracking, active headcount reconciliation.
HR Business Partners	Primary Users	Department-level analysis, attrition drivers, and employee performance splits.
Finance Controller	Reviewer	Payroll cost accuracy, leave liability, and salary comparison across departments.
Executive Management	Consumers	Headline KPIs, high-level trends, and overall turnover rates.
IT / Data Team	Data Provider	Data quality, refresh reliability, and workspace access control.
BI Analyst	Developer	Data model, DAX calculation accuracy, and BRD compliance.

6. DATA SOURCES AND DATA GRAIN

Fact_HR holds one row per employee record (key `Fact_ID`, business key `Employee_ID`) with numeric facts and foreign keys to each dimension table (`Dim_Department`, `Dim_JobRole`, `Dim_Location`, `Dim_Education`, `Dim_MaritalStatus`, `Dim_EmploymentStatus`, `Dim_Employee`, `Dim_Date`). The dataset is a point-in-time snapshot ending **28 December 2022**.

7. BUSINESS REQUIREMENTS

7.1 Page 1 - Workforce Overview

Purpose: current workforce size, cost, performance and workload (Headcount: 10,500, Avg Monthly Salary: 18,270 EGP, Avg Tenure: 5.0 yrs, Avg Performance: 3.33).

ID	Requirement	Priority
WO-01	Display total and active employees as headline cards (Total: 10,500 | Active: 8,941)	Must
WO-02	Show new hires, new hires YTD, prior-year comparison and YoY %	Must
WO-03	Show cumulative hires over time (2013-2022) and hires by year	Should
WO-04	Show headcount share by department (Sales, Marketing, Finance, Operations, IT, HR, Customer Success)	Must
WO-05	Show monthly and annual payroll, average salary (18,270 EGP) and median salary	Must
WO-06	Show average salary vs. company average by any dimension	Should
WO-07	Show average performance score (3.33), high performers % and rating distribution	Must
WO-08	Show training hours (total, average) and overtime hours (total, average)	Should
WO-09	Show average leave balance, total leave liability (days) and average tenure (5.0 yrs) with tenure-band distribution	Should
WO-10	Dynamic page title showing active employee count	Could

7.2 Page 2 - Demographics

Purpose: workforce composition and pay equity (Active Headcount: 8,941, Female %: 49.8%, Gender Pay Gap: 0.1%, Avg Age: 42.2 yrs).

ID	Requirement	Priority
DM-01	Male/female counts and percentages	Must
DM-02	Average salary by gender and Gender Pay Gap %	Must
DM-03	Average, youngest and oldest age; headcount by age group	Must
DM-04	Married employees count and %	Should
DM-05	Postgraduate (Master or PhD) % and headcount by degree level and major	Should
DM-06	Distinct offices and governorates; headcount by city/governorate	Should

ID	Requirement	Priority
DM-07	Dynamic page title with female % and average age	Could

7.3 Page 3 - Attrition

Purpose: measure, benchmark and explain employee turnover.

ID	Requirement	Priority
AT-01	Show leavers, voluntary leavers, involuntary leavers	Must
AT-02	Show attrition rate, voluntary and involuntary rates, and retention rate	Must
AT-03	Benchmark every slice against the company attrition rate (difference in percentage points)	Must
AT-04	Colour-code attrition (red above company average, green at or below)	Should
AT-05	Compare leavers vs. stayers on salary, performance, overtime and training	Must
AT-06	Show high-performer leavers and high-performer attrition rate	Should
AT-07	Show monthly salary lost to leavers	Could
AT-08	Analyse attrition by department, role, location, tenure band, age group, salary band and gender	Must
AT-09	Dynamic page title with overall attrition rate	Could

7.4 Cross-page requirements

7.4 Cross-Page Functional Business Requirements

ID	Requirement
GN-01	Global slicers synced across all report pages with a clear-filters option.
GN-02	Consistent visual layout theme and page navigation buttons.
GN-03	Bands and ratings sort in logical business order (via hidden sort columns), not alphabetically.
GN-04	All currency values shown in EGP with standard thousands separator.

8. KPI DICTIONARY

KPI	Definition
Total Employees	Distinct count of Employee_ID
Active Employees	Total Employees where Is Active = 1
New Hires / YTD / PY / YoY %	Employees by hire date; year-to-date; same period last year; relative change
Cumulative Hires	Running total of new hires up to the selected date
Headcount % of Total	Active employees / active employees across the current selection
Full-Time %	Active full-time employees / active employees
Monthly / Annual Payroll	Sum of active monthly salaries; monthly × 12
Avg / Median Monthly Salary	Average / median salary of active employees
Avg Salary vs Company	(Slice average − company average) / company average
Avg Performance Score	Average Performance_Score of active employees
High Performers %	Active employees with score ≥ 4 / active employees
Avg / Total Training Hours	YTD training hours, active employees
Avg / Total Overtime Hours	YTD overtime hours, active employees
Avg Leave Balance / Total Leave Liability	Average / sum of leave days, active employees
Avg Tenure (Years)	Average of (report date − hire date) / 365.25
Male % / Female %	Gender headcount / active employees

KPI	Definition
Gender Pay Gap %	(Male avg salary − Female avg salary) / Male avg salary
Avg / Youngest / Oldest Age	Age statistics, active employees (using Report Date 28 Dec 2022)
Married %	Married active employees / active employees
Postgraduate %	Active employees with Master or PhD / active employees
Leavers	Employees flagged Is Leaver
Voluntary / Involuntary Leavers	Leavers by Attrition Type
Attrition Rate	Leavers / Total Employees
Voluntary / Involuntary Attrition Rate	Respective leavers / Total Employees
Retention Rate	1 − Attrition Rate
Company Attrition Rate	Attrition rate with all dimension filters removed
Attrition vs Company (pp)	(Attrition Rate − Company Attrition Rate) × 100
High Performer Attrition Rate	High-performer leavers / all employees with score ≥ 4
Leavers Monthly Salary Lost	Sum of monthly salaries of leavers

9. FUNCTIONAL REQUIREMENTS (REPORT BEHAVIOUR)

- **FR-1 Fixed report date:** All age-at-date and tenure calculations use a single configurable Report Date measure (28 Dec 2022), matching the last date in the data, rather than today's date.
- **FR-2 Display folders:** Measures are grouped as 1 Workforce Overview, 2 Demographics, 3 Attrition, plus Titles, Settings and Flags.
- **FR-3 Interactivity:** Every visual cross-filters the others. Slicers sync across pages.
- **FR-4 Navigation:** A page navigator and bookmarks allow switching between pages and resetting filters.
- **FR-5 Conditional formatting:** Attrition visuals use the Attrition Color measure; variance KPIs show sign and colour.
- **FR-6 Export:** Users can export summarised data from visuals where permitted.

10. NON-FUNCTIONAL REQUIREMENTS

Category	Requirement
Performance	Pages load in under 5 seconds; visual interactions respond in under 2 seconds on the full dataset
Accuracy	KPIs reconcile to source data with zero variance on headcount and payroll
Security	HR data is confidential. Restrict workspace access; apply row-level security by department for HR Business Partners if required; salary visible only to authorised roles
Usability	Maximum of 3 pages, consistent layout, accessible colours (not red/green alone), clear titles and tooltips
Maintainability	Measures defined once and reused; model documented; naming conventions followed (Dim_/ Fact_ prefixes)
Compatibility	Power BI Desktop / Service; desktop and tablet layouts
Refresh	Scheduled refresh (monthly recommended); the Report Date updated at each refresh

11. ASSUMPTIONS, CONSTRAINTS AND DEPENDENCIES

Assumptions:

- The data is a snapshot as of 28 Dec 2022; one row per employee in Fact_HR.
- Employment Status values include Active and the leaver statuses mapped in Section 13.
- Monthly_Salary_EGP is the current gross monthly salary; Performance_Score is on a 1-5 scale.

Constraints:

- Time analysis is anchored to hire date only; there is no termination date, so attrition cannot be trended by exit month.
- The model is a snapshot, so headcount-over-time and monthly turnover rates cannot be reproduced historically.

Dependencies:

- Access to source data and confirmation of status values by HR.
- HR sign-off on KPI definitions and performance-rating thresholds.

12. DATA QUALITY AND VALIDATION REQUIREMENTS

- Unique Employee_ID and Fact_ID; no orphan foreign keys in any relationship.
- Hire_Date_ID must be a valid YYYYMMDD value and fall on or before the report date.
- No negative salary, hours or leave values; Performance_Score between 1 and 5.
- Dim_Date must cover every hire date.
- Reconciliation: total employees = active + leavers (+ other statuses).

13. BUSINESS RULES

Rule	Definition
BR-1 Active	Employment status = Active
BR-2 Leaver	Status is a resignation or termination value (Resigned, Resignation, Quit, Voluntary, Terminated, Termination, Fired, Involuntary)
BR-3 Voluntary	Resigned, Resignation, Quit, Voluntary
BR-4 Involuntary	Terminated, Termination, Fired, Involuntary
BR-5 High performer	Performance_Score ≥ 4
BR-6 Rating bands	Outstanding ≥ 4.5; Exceeds ≥ 3.5; Meets ≥ 2.5; otherwise Needs Improvement
BR-7 Tenure	(Report Date − Hire Date) / 365.25, in years
BR-8 Payroll	Active employees only; annual = monthly × 12
BR-9 Gender pay gap	Positive value means men earn more than women on average

14. RISKS AND ISSUES IDENTIFIED IN THE CURRENT MODEL

#	Risk / issue	Impact	Recommended action
R1	Age uses TODAY(), while tenure uses the fixed report date	Ages, age groups and averages are inconsistent with tenure and drift over time	Recalculate age using the Report Date (28 Dec 2022)
R2	Leaver status list contains many guessed spellings	Mis-classified leavers if actual values differ	Confirm exact Status values with HR and keep the list aligned
R3	Attrition Rate = Leavers / Total Employees (all statuses), while other metrics use active employees	Possible confusion between headcount bases	Document the denominator; add average-headcount basis if needed
R4	Avg Salary Stayers is defined as active employees	Correct only if every non-leaver is Active	Validate against the status list
R5	Only one relationship to Dim_Date (hire date)	Time trends reflect hiring, not exits or active headcount	State this clearly on the report; request termination date from source
R6	Calculated columns on the fact table use RELATED()	Slower refresh on larger data	Move flags to Power Query or source if data grows
R7	Sensitive salary and personal data	Privacy and compliance exposure	Apply role-based access and RLS; anonymise detail views
R8	Unused auto date/time tables (LocalDateTable_, DateTableTemplate_) in the model	Model bloat	Disable Auto date/time in options

15. ACCEPTANCE CRITERIA

1. All three pages are delivered and every requirement marked Must is met.
2. KPI values reconcile with an independent SQL or Excel calculation (tolerance: zero for counts, 0.01% for percentages).
3. All slicers and cross-filtering work across pages with no blank or erroneous visuals.
4. Bands and ratings sort logically; formats match the KPI dictionary.
5. Report Date is 28 Dec 2022 in all age and tenure results.
6. Performance targets in Section 10 are met.
7. Business Owner signs off after a user acceptance test (UAT).

16. DELIVERY PLAN AND MILESTONES

Phase	Activities	Output
1. Requirements	Stakeholder workshops, KPI definitions	Signed-off BRD
2. Data modelling	Star schema, relationships, calculated columns	Validated semantic model
3. Measures	DAX/TMDL measures in display folders	KPI library
4. Report design	Wireframes, three pages, theme, interactions	Draft report
5. Testing and UAT	Data reconciliation, user testing	Test report, defect log
6. Deployment	Publish, access control, refresh schedule	Live report
7. Handover	User guide, documentation, training	Handover pack

17. FUTURE ENHANCEMENTS

- Add termination date to enable monthly turnover trends and tenure-at-exit analysis.
- Headcount snapshots by month for true trend reporting.
- Predictive attrition risk scoring.
- Compensation benchmarking (compa-ratio) and succession planning views.

18. GLOSSARY

Term	Meaning
YTD / PY / YOY	Year-to-date / Prior year / Year over year
RLS	Row-level security
pp	Percentage points